

TORSIONVERSE SERIES 3: JOBSON CELL COHERENCE —

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DYNAMIC RIGIDITY, LOSSLESS PROPAGATION, AND THE HIGGS AS UV CUTOFF

Predecessor: docs/series1/doc_jobson_cell.txt (DOI 10.5281/zenodo.22032906)

Companion scripts: analysis/demos/cell_coherence_doc.py [CC1-CC10, 10/10 PASS]

analysis/quantum/cell_cycle_corpuscle.py	[CC_G1-CC_SYNC_2, 12/12 PASS]
analysis/quantum/cell_cycle_stability.py	[STAB_G1-STAB_T3, 7/7 PASS]
analysis/quantum/corpuscle_force_balance.py	[CF1-CF6, 7/7 PASS]
analysis/quantum/gluon_phase_sync.py	[GPS1-GPS6, 6/6 PASS]
analysis/quantum/gluon_bode_stability.py	[BS1-BS5, 5/5 PASS]
analysis/quantum/maxwell_nexus_connection.py	[MN1-MN7, 7/7 PASS]
analysis/quantum/tau_natural_path.py	[TNP1-TNP3, 3/3 PASS]

ABSTRACT

The Jobson cell (I_h icosahedron, Series 1) is treated as a rigid unit in all Series 1 and Series 2 derivations. This paper provides the quantitative justification: (1) the bilateral tau (I52) synchronizes all 20 face-center gluon maxima in $5.39 \times L_J/c = 1.79 \times 10^{-25} \text{ s}$ — shorter than ANY physical interaction timescale at sub-E_cell energies; (2) the Maxwell-critical lattice (3V-E=6) has zero-energy soft modes and cannot absorb wave energy as heat; (3) both results break down at exactly $E_{\text{cell}} = 124.8 \text{ GeV}$ — the Nyquist condition where the photon wavelength equals L_J . At this frequency, the wave couples to the cell's A_g (Higgs) mode rather than propagating through it. The Higgs mass IS the UV cutoff of the torsion medium, derived from wave mechanics rather than spontaneous symmetry breaking alone. Three consequences follow: lossless EM propagation below E_{cell} , dynamic rigidity for all nuclear and atomic processes, and a clean boundary above which cell-scale physics is directly accessible. Exceptions (Pauli-locked states, Zone-1-excited configurations) are noted as open items.

SECTION 1: THE COHERENCE QUESTION

Series 1 treats each Jobson cell as a rigid unit — a solid icosahedral object that rotates, translates, and locks with neighbours via the co-rotation chain (cell_rotation_propagation.py RP1-RP4). This is the "geared ball" picture.

The cell's structural modes are built from three classes of corpuscle. The gluons (2G_g) are STANDING oscillations anchored at each of the 30 edges — three per triangular face, together constituting that face [GH0c, FG3-FG8].

The muon (G32) is a TRAVELING wave through the edge network, using the gluon-defined face boundaries as

waveguides. The tau (I52) makes a Hamiltonian circuit of all 20 face-center nexuses (at inradius = $0.756 \times L_J$ on the outer surface). Between nexuses the tau travels as a straight-line CHORD through the cell interior — not a surface-following path. At mid-hop the chord dips to $r = 0.706 \times L_J$, which is $0.103 \times L_J$ inside the edge midpoint ($r_{\text{mid}} = 0.809 \times L_J$); the path passes beneath the edge, so no surface-clearing is required. Each hop length is $\phi/3 \times L_J \approx 0.54 \times L_J$ [GH3]. At each face-center nexus all three gluons of that face reach their standing-wave amplitude maximum simultaneously [GH0b] — the tau arrives from the interior, scatters off this inward-facing gluon maximum, and deflects by 72° (C5, same as the muon at vertices) [GH2, GH5]. The 138.19° dihedral (JC5) is the fold of the cell surface at the edge each chord crosses — geometry of the shell, not the tau's angle. "Outer surface only" refers to the NEXUSES (all 20 at inradius); the path segments pass through the interior. No inward-spiraling tau path exists [CE1-CE5 — a separate topology where tau winds toward $r=0$, forbidden]. When an external force is applied — say, a proton's Zone 3 co-rotation locking a cell face — all these internal modes must receive and transmit the perturbation coherently. If they don't, the cell "flexes" rather than rotates as a unit, and the rigid-body treatment breaks down.

The question is not whether the cell is in static equilibrium (proved by the zero-force checks at all 62 nexuses, `jobson_cell_force_balance_vectors.py` G3D1-G3D8). The question is whether the cell responds DYNAMICALLY as a rigid unit when external forces are applied at any sub- E_{cell} energy.

This paper proves it does, and identifies E_{cell} as the exact energy where the rigid-body picture ends.

SECTION 2: DYNAMIC RIGIDITY — THE TIMESCALE ARGUMENT [CC1-CC3]

The cell's internal coherence is maintained by the bilateral tau (I52) — TWO corpuscle photons traversing the complete Hamiltonian circuit of all 20 face-center nexuses in opposite directions [GH5]. At each nexus all three gluons of that face reach their standing-wave amplitude maximum simultaneously [GH0b], so each tau hop visits one face's full gluon state. Each photon makes 10 hops before meeting its partner at the antipodal nexus (hop 10); together they have visited all 20 face-center gluon maxima. Hop length = $\phi/3 \times L_J$ [GH3]. The full-cell gluon synchronization time is:

$$t_{\text{tau_sync}} = 10 \times (\phi/3 \times L_J)/c = 5.39 \times L_J/c = 1.79 \times 10^{-25} \text{ s} \quad [\text{CC1}]$$

Compare to physical interaction timescales:

Strong force at r_p :	$t_{\text{strong}} = r_p / c = 2.80 \times 10^{-24} \text{ s} = 15.7 \times t_{\text{tau_sync}}$
EM at nuclear scale:	$t_{\text{EM}} = r_p / (c \cdot \alpha) = 3.84 \times 10^{-22} \text{ s} = 2,150 \times t_{\text{tau_sync}}$
Atomic scale:	$t_{\text{atom}} \sim 10^{-17} \text{ s} = 10^8 \times t_{\text{tau_sync}}$

For EVERY physical interaction below E_{cell} , the bilateral tau has visited all 20 face-center gluon maxima before the external force can change meaningfully. The cell therefore responds as a coherent unit for all sub- E_{cell} interactions.

The co-rotation chain (RP1-RP4) confirms zero attenuation: each handoff is exactly 72° (C5 geometry), and the muon's pole-to-pole zigzag [muon_symmetry.py MS1-MS7] delivers the signal to all 12 vertices equally (35 circuits/vertex over the 70 distinct circuits). No corpuscle is left behind.

The tau-lever synchronization is Bode-stable: a gluon phase offset δ at a nexus generates a restoring force $\propto (1 - \cos \delta)$ [GPS1-GPS4], giving second-order (δ^2) stability. The per-circuit loop gain is $G_{\text{total}} = 3.75 \times 10^6$ — a $2,670\times$ damping factor per tau traversal [BS1-BS4]. Any phase error is suppressed within a single circuit. [GPS1-GPS6, BS1-BS5]

CONCLUSION: The "geared ball" treatment is JUSTIFIED for all physical processes below E_{cell} . The rigid-body approximation is not an assumption — it follows from the bilateral tau synchronizing all face gluons in 1.79×10^{-25} s, which is $15.7\times$ shorter than the strong force timescale. [CC2]

SECTION 2a: CORPUSCLE CYCLE TIMING AND PERTURBATION RECOVERY [CC_G1-CC_T4, STAB_G1-STAB_T3]

The three corpuscle types have distinct cycle periods, verified without wave mathematics by cell_cycle_corpuscle.py and cell_cycle_stability.py:

```
Gluon (2 counter-propagating corpuscles per edge, 60 total):
  Period = 2 edge-crossings – corpuscle bounces between two antipodal
  vertices; the two corpuscles per edge are always at opposite vertices
  (standing-wave pattern). [CC_G1, CC_G2, CC_G3]

Muon (G32, 20 bilateral corpuscles = 10 orbit-A circuits):
  Period = 6 edge-crossings – one full pole-to-pole zigzag circuit.
  The backward corpuscle is always 3 steps behind the forward (bilateral
  = half-circuit offset). [CC_M1, CC_M2, CC_M3]

Tau (I52, 2 bilateral corpuscles):
  Period =  $20 \times (\phi/3) \times (L_J/c)$  – Hamiltonian circuit of all 20
  face-center nexuses. Bilateral corpuscles meet at step 10 (antipodal
  face center). [CC_T1, CC_T2, CC_T4]
```

Gluon and muon periods are commensurable (muon = $3 \times$ gluon). The tau period is INCOMMENSURABLE with both — ϕ is irrational, so the tau never returns to the same relative gluon phase, providing aperiodic sampling of all 20 gluon states. [CC_SYNC_1, CC_SYNC_2]

2a.1 Tau perturbation recovery (gluon-guided backtracking)

If the tau is deflected to a wrong face-center nexus (1-face perturbation at cycle 5), it does NOT remain permanently out of phase. The gluon amplitude maxima at each nexus [GH0b] guide the recovery:

- From the wrong face, the tau continues its natural 72° deflections.

GEOMETRIC RESULT [TNP1-TNP3]: both non-incoming neighbors are always at exactly 72° (dodecahedron vertex symmetry, TNP1). A blind consistent direction would orbit a 5-face pentagon (dodecahedral ring, TNP2) -- so the gluon amplitude at each nexus actively selects between the two equal-angle options. With correct gluon selection, a full 20-face Hamiltonian circuit always exists from any wrong-face start (TNP3).

- All 20 face centers are recovered in 18 extra steps (STAB_T1).
- Recovery is efficient: 18 steps for 19 remaining faces (STAB_T2).
- After recovery, normal Hamiltonian cycling resumes from cycle 6 onward with no residual error (STAB_T3).

The muon is robust by construction: any cyclic phase shift leaves the 6-vertex circuit unchanged (STAB_M1, STAB_M2). The gluon period = 2 is preserved under any phase swap (STAB_G1, STAB_G2).

CONCLUSION: All three corpuscle types return to full structural coverage after a single-nexus perturbation. The tau continues its natural 72° deflections from wherever it lands — no active rerouting required. The icosahedral geometry guarantees full face-center coverage within 18 steps of the perturbation.

SECTION 3: LOSSLESS PROPAGATION — THE MAXWELL CRITERION [CC4-CC6]

For a force to permanently alter the cell's internal configuration, it would need to scatter energy INTO the cell lattice. This is forbidden by the Maxwell criticality of the Jobson cell ($3V-E = 6$, SY1):

3.1 Zero-energy soft modes

The icosahedral dynamical matrix has exactly 6 zero eigenvalues (SY11): the T_{1g} and T_{2g} soft modes. Zero eigenvalue means zero frequency = zero energy cost to propagate. T_{1g} and T_{2g} waves store NO energy in the lattice bonds and experience no resistance. A force applied in a soft-mode direction passes THROUGH the cell without depositing energy. [CC4]

3.2 No scattering channel at long wavelength

Maxwell criticality ($3V-E = 6$) places the lattice at the exact rigidity transition. At this point, the soft modes (T_{1g} , T_{2g}) are DECOUPLED from the hard-bond modes at long wavelength ($k \rightarrow 0$). There is no scattering channel from the propagating soft modes into the constrained hard bonds. [CC5]

The gluon confinement underlying this decoupling is an algebraic identity: $\chi(2G_g, C5) = -1$ (phase flip at every vertex) and $\chi(2G_g, C3) = +1$ (constructive at face centers) follow directly from $n=5$ edges/vertex and Maxwell criticality alone — no dynamical assumption required. Gluons are edge-confined; T_{1g} and T_{2g} are not; this is why the soft modes decouple. [MN1-MN7]

3.3 Thermal energy cannot excite cells

The cell energy scale is $T_{\text{cell}} = E_{\text{cell}} / k_B = 1.45 \times 10^{15}$ K. The hottest known matter (QGP, $\sim 2 \times 10^{12}$ K) is $0.0014 \times T_{\text{cell}}$. No physical environment

in the observable universe can thermally excite a Jobson cell. The cells stay in their ground state; no absorption channel is thermally accessible. [CC6]

CONCLUSION: Below E_{cell} , T_{1g} and T_{2g} waves propagate without absorption. Heat IS the disordered T_{1g} wave field. The medium is a cold, lossless substrate. This is NOT a separate postulate — it follows from the geometry of the I_h cell.

(Scope condition: Pauli-locked states and Zone-1-excited configurations alter the effective dynamical matrix. Dynamic rigidity and lossless propagation within those states are Series 3 open items. See Section 6.)

SECTION 4: THE NYQUIST CONDITION — WHERE RIGIDITY ENDS [CC7-CC9]

The rigid-body and lossless-propagation proofs are valid for ALL sub- E_{cell} interactions. What happens AT E_{cell} ?

The Nyquist condition (SY6, `torsionverse_doc.py`) states:

$$E_{\text{Nyquist}} = \hbar \times (2\pi c / L_J) = 124.8 \text{ GeV} = E_{\text{cell}} \quad [\text{CC7}]$$

This is the energy at which one full EM oscillation has a spatial period exactly equal to L_J . At this energy, the wave can NO LONGER average over multiple cells — it resolves the discrete lattice.

The TEMPORAL version of the same condition (SY20b):

$$T_{\text{Nyquist}} = 2\pi \hbar / E_{\text{cell}} = L_J / c = 3.31 \times 10^{-26} \text{ s} \quad [\text{CC8}]$$

The photon period AT E_{cell} equals the single-corpuscle cell-edge transit time (L_J/c). The cell can no longer respond before the next oscillation cycle — the coherence condition breaks down.

Both the spatial (wavelength = L_J) and temporal (period = transit time) Nyquist conditions give identical energies, confirming that E_{cell} is the single UV cutoff of the torsionverse medium. [CC9]

THE THREE ENERGY REGIMES:

- $E \ll E_{\text{cell}}$: Medium is continuous. T_{1g}/T_{2g} propagate as smooth waves. Cell is dynamically rigid. No absorption. EM and gravity described by T_{1g} and T_{2g} long-wavelength modes.
- $E = E_{\text{cell}}$: Nyquist boundary. Wave resolves individual cells. Couples to A_g (isotropic bulk mode = Higgs). Energy goes INTO the cell structure, not through it. SSB / jamming.
- $E > E_{\text{cell}}$: Wave sees cell discreteness. Can create new cells (cloning requires $m_{\text{crit}} = 9.933 \text{ GeV}$; full cell creation requires E_{cell}). Above E_{cell} : medium becomes opaque to that wave.

SECTION 5: HIGGS AS UV CUTOFF — THE A_g CONNECTION [CC10]

The A_g irreducible representation is the totally symmetric (isotropic) mode of I_h . From `jobson_cell_doc.py` J13:

$$T_{1g} \times T_{1g} \rightarrow A_g \text{ appears exactly once (unique scalar combination)}$$

A_g is the Higgs field of the torsionverse (J7, J20, J21). Its mass is the energy at which the I_h lattice reaches jamming (Maxwell criticality = SSB): E_{cell} = 124.8 GeV. This is established in Series 1.

This paper adds the wave mechanics perspective: A_g is also the mode excited AT the Nyquist boundary. When a wave with E = E_{cell} hits a cell, it cannot continue as a T_{1g} propagating wave — it couples to the unique scalar A_g mode. The energy goes into the isotropic bulk compression, which is jamming/SSB.

The Higgs mass is therefore the UV cutoff FROM TWO INDEPENDENT PERSPECTIVES:

- (1) Algebraic: A_g = jamming mode, mass = E_{cell} (J7, J20, Series 1)
- (2) Wave-mechanical: A_g = excitation at Nyquist boundary (this paper)

These are the same condition stated in two different languages. The Higgs mass is not fine-tuned — it is the frequency at which the elastic icosahedral medium transitions from a continuous propagating-wave substrate to a discrete cell-interacting regime. [CC10]

NOTE ON THE HIERARCHY PROBLEM:

In standard QFT, the Higgs mass at ~125 GeV is unnaturally small compared to the Planck scale (~10¹⁹ GeV) — the hierarchy problem. In the torsionverse, there is no hierarchy problem because the Higgs mass is DERIVED: it is L_J/c in natural units, set by the icosahedral cell geometry (alpha, phi, r_p). The "problem" does not arise because the Higgs is not a fundamental scalar with arbitrary mass — it is the A_g mode of the I_h cell lattice.

SECTION 6: EXCEPTIONS — SCOPE OF THE RIGIDITY PROOF

The dynamic rigidity (Section 2) and lossless propagation (Section 3) proofs apply to the GROUND-STATE medium. Two configurations are outside this scope:

6.1 Pauli-locked state (Mode 3 of spinning pyramid, doc_series2_spinning_pyramid.txt)

When cells are all in the same Hopf winding state, the effective dynamical matrix changes. The zero-eigenvalue argument (Section 3.1) may not hold — the locked cells have a different soft-mode structure. Whether a Pauli-locked region remains lossless and rigid is an open question. [Series 3 target]

6.2 Zone-1-excited configuration (strange quark G_u fluctuation, CA28b)

A transient strange quark (G_u, m_{strange} ≈ m_p/2) entering Zone 1 locally modifies the cell's chi coupling. The soft-mode decoupling may temporarily fail in that region, enabling local energy absorption. This may be relevant to the proton's stability — the cell configuration around Zone 1 may self-adjust to remain non-absorbing in the ground state while allowing a transient absorbing path for excited states. [Series 3 target]

6.3 E+ vertex continuation rule

The (1,2) Hopf winding of the electron is topologically unique (J20, V7-V8). The tau (I52) and muon (G32) bounce patterns are explicitly proven stable (CG9, CG17: 72° deflection at all steps). For the E+ mode, no script currently proves which of the 5 outgoing edges at each vertex the winding selects. The I_h symmetry implies uniqueness; the explicit geometric proof is a Series 3 target.

SECTION 7: CONCLUSIONS

Three results establish the physical basis for treating Jobson cells as rigid, lossless units in all sub-E_{cell} physics:

1. DYNAMIC RIGIDITY [CC1-CC3]: bilateral tau sync time ($5.39 \times L_J/c = 1.79 \times 10^{-25}$ s) is shorter than the strong force timescale by 15.7× and EM by 2,150×. All face gluons synchronized; cell responds as a rigid unit.
2. LOSSLESS PROPAGATION [CC4-CC6]: Maxwell criticality (3V-E=6) decouples soft modes from hard bonds. T_{cell} = 1.45×10^{15} K is unreachable at any physical temperature. No absorption channel exists. Heat is disordered T_{1g} wave energy, not thermal excitation of the medium.
3. HIGGS AS UV CUTOFF [CC7-CC10]: E_{cell} is the Nyquist boundary where both the spatial ($\lambda = L_J$) and temporal ($T = L_J/c$) conditions are satisfied. At this energy, the wave couples to A_g (Higgs) rather than propagating. The Higgs mass is derived, not fine-tuned; the hierarchy problem dissolves.
4. CORPUSCLE CYCLE STABILITY [CC_G1-CC_T4, STAB_G1-STAB_T3]: all three corpuscle types recover from a single-nexus perturbation without residual error. Tau-lever Bode loop gain G_{total} = 3.75×10^{11} (2,670× damping per circuit) confirms the synchronization in result 1 is actively stabilized, not a passive timescale bound. [GPS1-GPS6, BS1-BS5]

Companion scripts: analysis/demos/cell_coherence_doc.py [CC1-CC10, 10/10 PASS]

```
analysis/quantum/cell_cycle_corpuscle.py [CC_G1-CC_SYNC_2, 12/12 PASS]
analysis/quantum/cell_cycle_stability.py [STAB_G1-STAB_T3, 7/7 PASS]
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analysis/quantum/tau_natural_path.py [TNP1-TNP3, 3/3 PASS]
```

Cross-references: doc_jobson_cell.txt (J3, J7, J13, J20, J21, SY6, SY11)

```
torsionverse_doc.py (SY19a-c, SY20a-b)
cell_rotation_propagation.py (RP3-RP4)
```

EXTERNAL INPUTS: none. All results derived from E_{cell}, L_J, alpha, phi, r_p.